

A Monolithic Gas Pixel Detector with on On-Chip Micro-Gap Amplification

Luca Baldini

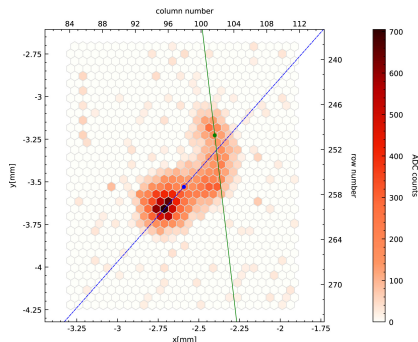
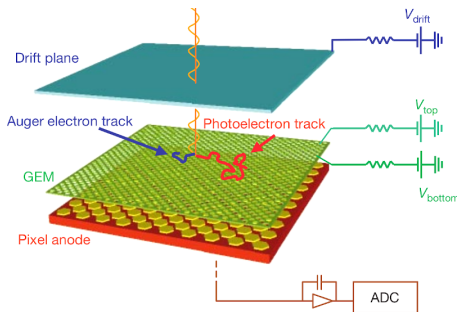
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9th International Conference on Micro Pattern Gaseous Detectors

Context 1/2

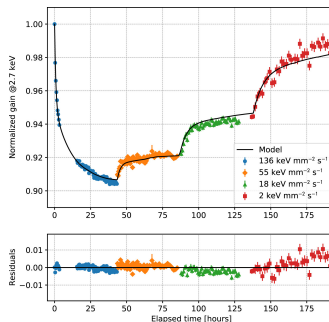
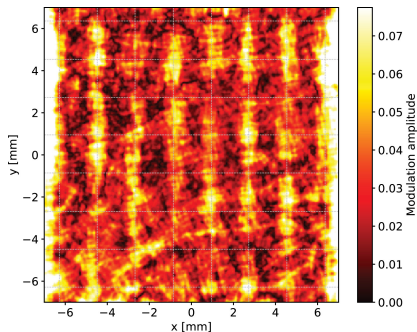
The Gas Pixel Detectors onboard the Imaging X-ray Polarimetry Explorer



- ▷ Imaging X-ray Polarimetry Explorer (IXPE)
 - ▷ NASA/ASI mission, launched December 2021, in operation
 - ▷ Entirely devoted to polarimetry in the 2–8 keV band
 - ▷ Three identical telescopes (mirrors + focal-plane detectors)
- ▷ Gas Pixel Detectors (GPDs)
 - ▷ GEM + custom readout ASIC to image the photoelectron track
 - ▷ Polarization is recovered via azimuthal distribution of the tracks

Context 2/2

Lessons learned: L. Baldini et al., Astroparticle Physics, 133 (2021)



- ▷ Spurious modulation (left)
 - ▷ Azimuthal response for unpolarized radiation not perfectly flat
 - ▷ Variations across the active area, mapping the GEM manufacturing
- ▷ Charging (right)
 - ▷ Gain variations due to charge deposition on the GEM dielectric
- ▷ Both effects in at least three different types of 50 μm -pitch GEM
 - ▷ Meeting the science goals required complex and long calibrations

And now: a blast from the past

The Micro Gap Chamber (MGC), circa 1993

Nuclear Instruments and Methods in Physics Research A 335 (1993) 69–77
North-Holland

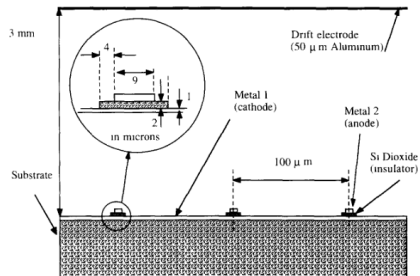


Fig. 1. A cross section of the detector internal structure.

The micro-gap chamber

F. Angelini, R. Bellazzini, A. Brez, M.M. Massai, R. Raffo, G. Spandre and M.A. Spezziga

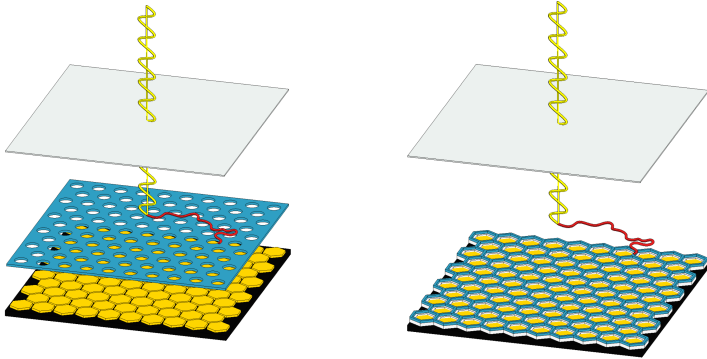
INFN-Pisa and University of Pisa, Via Livornese 582-I-56010 S. Piero a Grado, Pisa, Italy

Received 12 May 1993

The micro-gap chamber (MGC), a new type of positron sensitive proportional gas counter, is introduced. The device is built using microelectronics technology. In this detector the separation between the electrodes collecting the avalanche charge (the anode–cathode gap) is only a few microns. The time it takes to collect the positive ions is therefore very short (≈ 10 ns). The speed of the device now equals that of solid state detectors but it is more than three orders of magnitude higher than in standard proportional counters and one order of magnitude higher than in the recently introduced microstrip gas chamber (MSGC). As a result, the rate capability is extremely high ($> 9 \times 10^6$ c/mm² s). The amplifying electric field around the thin anode microstrip extends over a small volume but is very intense (270 kV/mm). It provides a gas gain of 2.5×10^3 at 400 V with 14% (FWHM) energy resolution at 5.4 keV. The anode pitch is 100 μ m and the readout is intrinsically two-dimensional. Because there is practically no insulating material in view, charging was not observed even at the highest rate. This device seems very well suited for instrumentation of the tracking system at the new hadron colliders (LHC/SSC) as well as in many other fields of research.

The driving idea

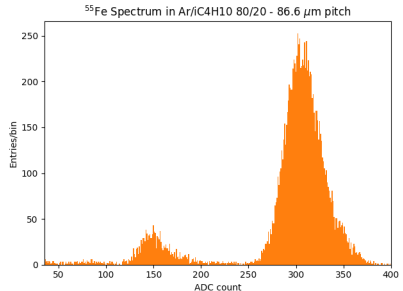
A tale of many questions



- ▷ Integrate the gain stage directly onto the readout chip
 - ▷ In such a way that it can be done entirely in a standard CMOS process
 - ▷ Tailored to applications requiring moderate $O(100)$ gain
- ▷ Complex enough that it cannot be realistically done in one step
 - ▷ Proof of concept: post-process an existing chip (XPOL)
 - ▷ Slight complication: XPOL is designed to read electrons, not ions

Q1: Can we do it at all? Yes.

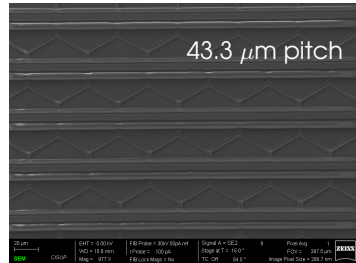
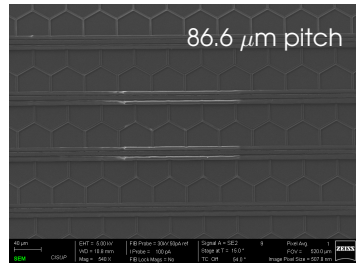
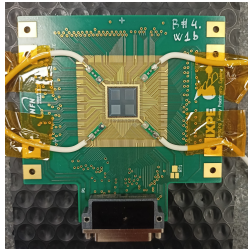
Rehearsing the printing technology 30+ years later



- ▷ Four $5 \times 5 \text{ mm}^2$ of structures deposited on a dummy Si substrate
 - ▷ Strips with 43.4 μm and 86.6 μm pitch (1 $\times$ and 2 $\times$ XPOL)
 - ▷ Two-dimensional hexagonal structure matching the XPOL pixels
 - ▷ Al (2 μm thick and 5 μm wide) on SU-8 (3–4 μm thick and 13 μm wide)
- ▷ Electrical and functional characterization (gain/energy resolution)
 - ▷ Substrate at HV, structures to spectroscopic preamplifier + MCA
 - ▷ Drift electrode at 1 cm distance, ⁵⁵Fe source
 - ▷ No position information—only pulse-height spectra
 - ▷ Note: noise causing a measurable contribution to the resolution

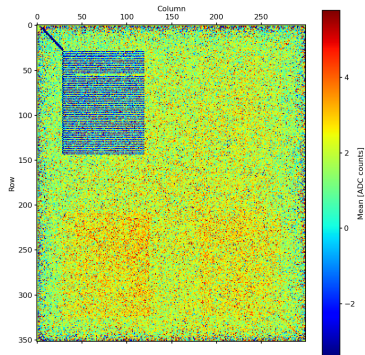
Q2: Can we do it on the chip? Yes!

Same manufacturing process, on a harsher substrate



Q3: Does the chip survive post-processing? Yes!

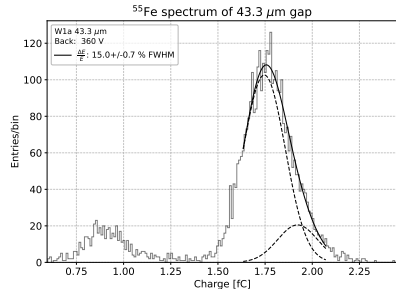
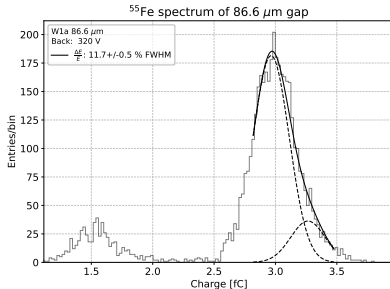
If we break the chip, there is no use in printing. . .



- ▷ Use the XPOL calibration system to measure pedestals and noise
 - ▷ One structure floating, all the other three grounded
- ▷ No significant change in pedestal and noise distributions
- ▷ Slight offset for the pixels immediately below the structures
 - ▷ And more so for the floating structure
 - ▷ **Nice tomography of the structure**—can you see the broken strip?

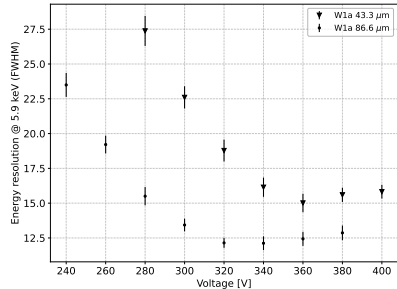
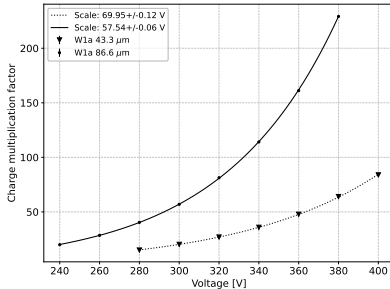
Q4: Do we get gas gain on chip? Yes!

Now things get interesting—we have a real viable detector



- ▷ Measurement of the gain and energy resolution *on chip*
 - ▷ Same setup of the dummy wafers: one spectroscopic channel + MCA
 - ▷ Reading the strips, as the chip has the wrong polarity!
 - ▷ (All XPOL pads connected to HV)
- ▷ Resolution below 12% at 5.9 keV demonstrated at 86.6 μm pitch
 - ▷ Slightly worse ($\sim$ 15%) at 43.3 μm pitch

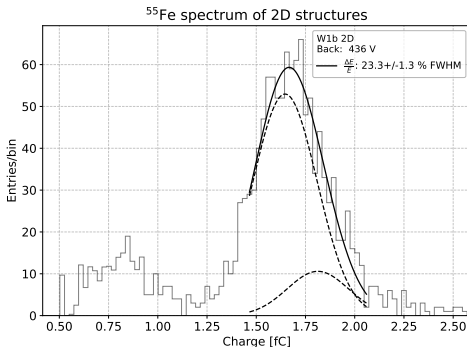
Basic characterization of the response



- ▷ Comprehensive characterization of the strip-like structures
 - ▷ Stable operation for several hours achieved in both cases
 - ▷ Target gain of ~ 100 reached at $\sim 330/400+$ V in the two cases
 - ▷ Excellent energy resolution at the working point

Q5: Does the 2-d version work? Yes!

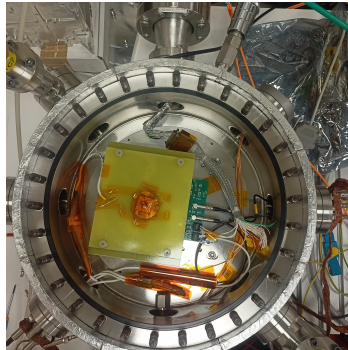
First demonstration of this kind of two-dimensional structures



- ▷ Significantly more challenging than the one-dimensional structures
 - ▷ Not surprising
 - ▷ Gain of 50+ demonstrated
 - ▷ Energy resolution not spectacular, but decent
- ▷ Additional work needed to achieve stable operation
 - ▷ Possible improvements to the design and manufacturing identified
 - ▷ (e.g., strip and pad geometry, passivation of some of the elements)

Q6: Does the chip withstand the avalanches? Yes!

This is effectively closing the loop



- ▷ Structures on chip tested in working condition in different setup
 - ▷ Readout chip connected to its back-end electronics
 - ▷ Structures connected to HV at the nominal working point
 - ▷ ^{55}Fe source, gas mixture, drift electrode, as in the previous tests
 - ▷ System not triggering due to wrong polarity
 - ▷ Continuous monitoring of the pixel response
- ▷ No damage observed after several hours of operation
 - ▷ Cannot read events, but we know we could with the right polarity

- ▷ New concept for a monolithic gas pixel detector with amplification on chip
 - ▷ Designed to be compatible with standard CMOS processes
 - ▷ Tailored for applications where moderate gain is sufficient
 - ▷ (X-ray polarimetry is a prime example, but not the only one)
- ▷ Proof of concept demonstrated via post-processing of an existing chip
 - ▷ Structures compatible with the XPOL readout chip successfully printed
 - ▷ Gain and energy resolution measured on chip
 - ▷ Two-dimensional structures demonstrated
 - ▷ No damage to the chip observed after several hours of operation
- ▷ Further work needed to optimize the design and manufacturing
 - ▷ Alternative structure geometries can be explored
- ▷ Next step: design a new chip with the right polarity
 - ▷ Most likely still via post-processing
- ▷ Final goal: going full out with a full monolithic CMOS design